

• Penalise variables in equations once only -1



SHENTON
COLLEGE

Year 11 Mathematics Applications (AEMAA)
Investigation 2 2019
Applying Linear Relationships

52

Name: SOLUTIONS

CALCULATOR ASSUMED MARKS: 52 TIME: 50 minutes

Question 1 [11 marks: 1,1,4,1,1,1,1,1]

Ruby is going to turn her jewellery-making hobby into a business. Her fixed costs are \$600 for equipment, tools etc and her variable cost for each necklace is \$15 for materials and other consumables.

(a) Write an equation which will enable her to determine the total cost \$C of making n necklaces.

$$C = 15n + 600$$

✓ Writes equation with correct variables

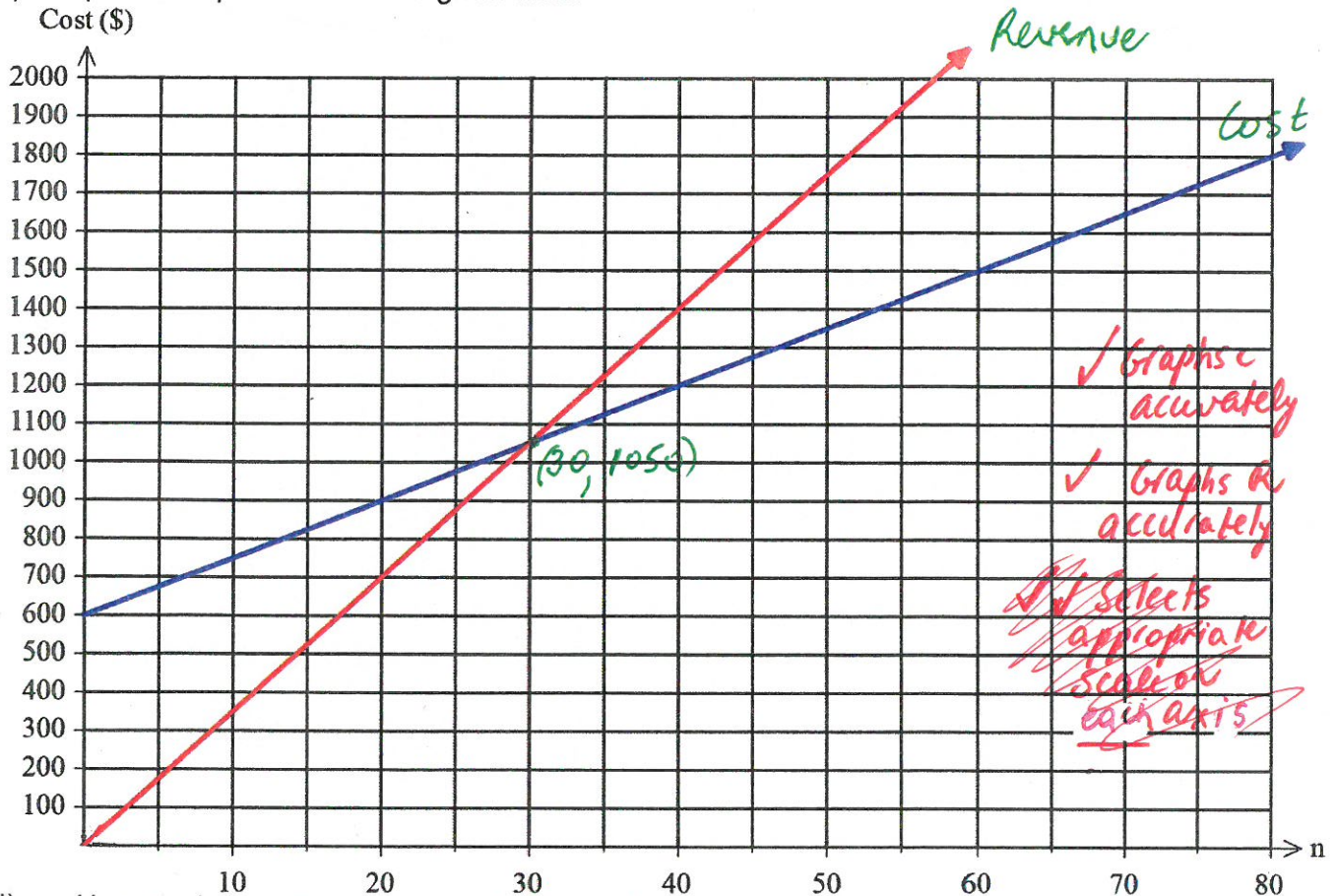
(b) She plans to sell her necklaces for \$35.

Write an equation which will enable her to determine the revenue \$R on selling necklaces. Revenue is the amount of money she will make from selling the necklaces.

$$R = 35n$$

✓ Writes correct equation with correct variables

(c) Graph both equations onto the given axes.



✓ Graphs C accurately

✓ Graphs R accurately

✓ Selects appropriate scale on each axis

(d) How much does it cost to make 40 necklaces?

\$1200

✓ Gives correct cost from equation or graph 5

(e) What is the revenue earned from selling 40 necklaces?

\$1400

✓ Gives correct Revenue from graph or equation

(f) What profit or loss is made from making and selling 40 necklaces?

\$200 Profit

✓ States \$200 - must state PROFIT.

(g) What is the break-even point?

(30, 1050)

✓ Gives correct point as co-ordinate.

(h) Explain what the break-even point means in this context.

Where the Revenue and Cost are equal meaning no profit or loss.

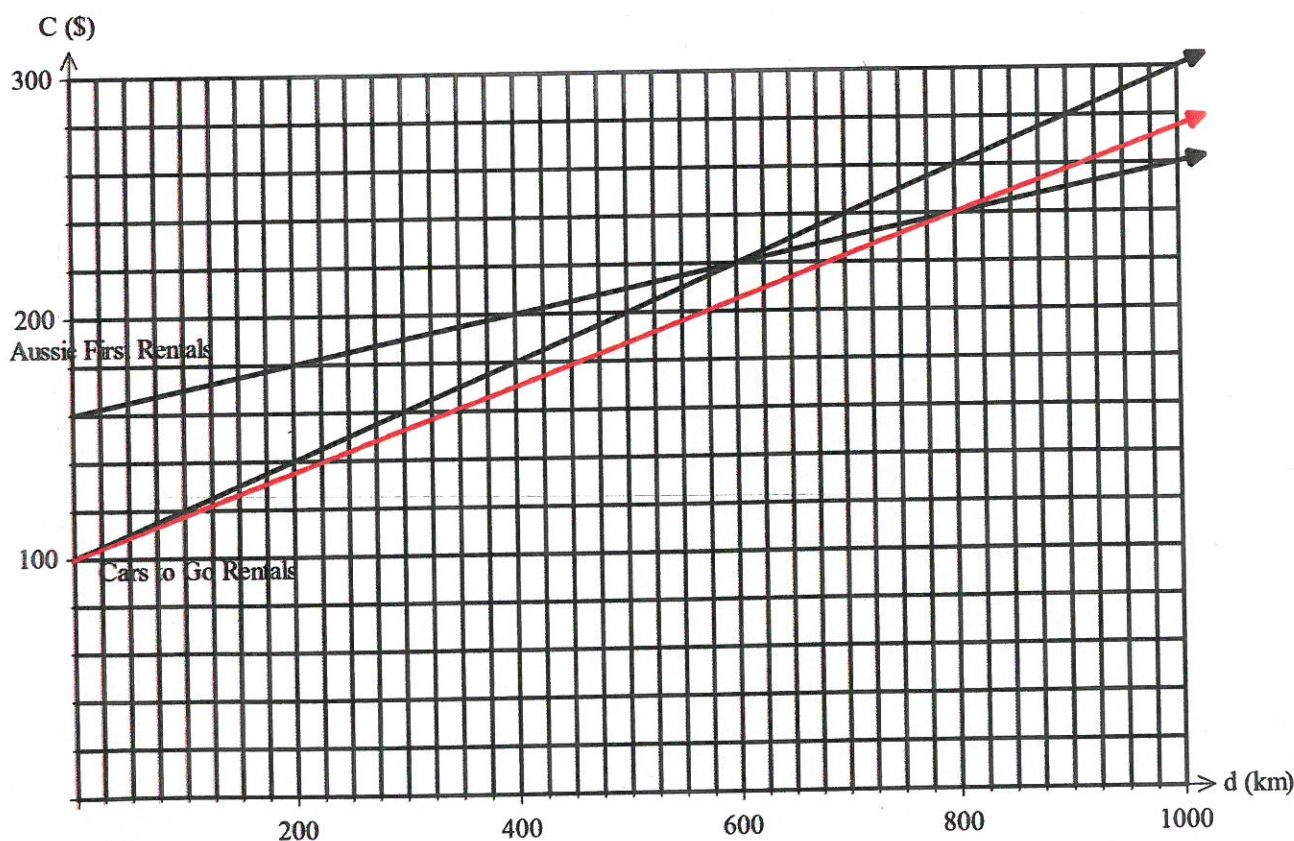
✓ Clear response given using Revenue and Cost OR no profit/loss

Question 2 [13 marks: 1,2,2,2,2,1,1,1,1]

Maxwell is considering hiring a 4WD for a camping trip to Geraldton. He researches the costs involved in the travelling to Geraldton and summarises his findings in the graph below.

The graph shows the rental cost \$C for travelling d kilometres for each of two companies:

Cars to Go Rentals and Aussie First Rentals.



(a) Determine the fixed hire cost for each of the companies.

Cars to Go Rentals \$100
Aussie First Rentals \$160

✓ States both fixed costs

- (b) Determine the cost per kilometre travelled for each company.

Cars to Go Rentals $\frac{220-100}{600-0} = 0.2$ \$0.2/km ✓
Aussie First Rentals $\frac{220-160}{600-0} = 0.1$ \$ 0.1/km ✓

Determines correct rates (accept cents)

- (c) Hence, or otherwise determine the equations for total rental cost for each company.

$$C_C = 0.2d + 100$$

$$C_A = 0.1d + 160$$

✓ Determines correct equations with correct variables*

- (d) Determine the point of intersection of the two lines and explain the significance of the point of intersection of the two lines.

$$(600, 220)$$

Both companies cost \$220 when 600 km travelled.

✓ States point of intersection

✓ Clear explanation using BOTH idea

- (e) Maxwells's home is 470 km from Geraldton. Maxwell only wishes to cost the first leg of the journey from his home (to Geraldton one-way). Which company should he choose and why?

Cars to Go Rentals.

Graph of Cars to Go Rentals below

the graph of Aussie First Rentals for first 600km of trip

✓ Gives correct company.

✓ Gives valid reason

Maxwell negotiates a $12\frac{1}{2}\%$ discount on the cost per kilometre with Cars to Go Rentals.

- (f) What is this discounted price per kilometre?

$$\$0.175$$

✓ Determines discounted rate.

- (g) Determine the linear equation for the discounted hire rate.

$$C = 0.175d + 100$$

✓ correct equation with correct variables.*

- (h) Add the graph of this equation to the graph above.

✓ Accurately graphs equation

- (i) Explain what this new graph tells about hiring 4 WD's in this situation.

The discounted price is cheaper for the Geraldton trip, but for travelling 800 km or more Aussie First Rentals would be cheaper.

✓ Clear explanation must mention $d \geq 800$ km

Question 3 [13 marks:4,3,5,1]

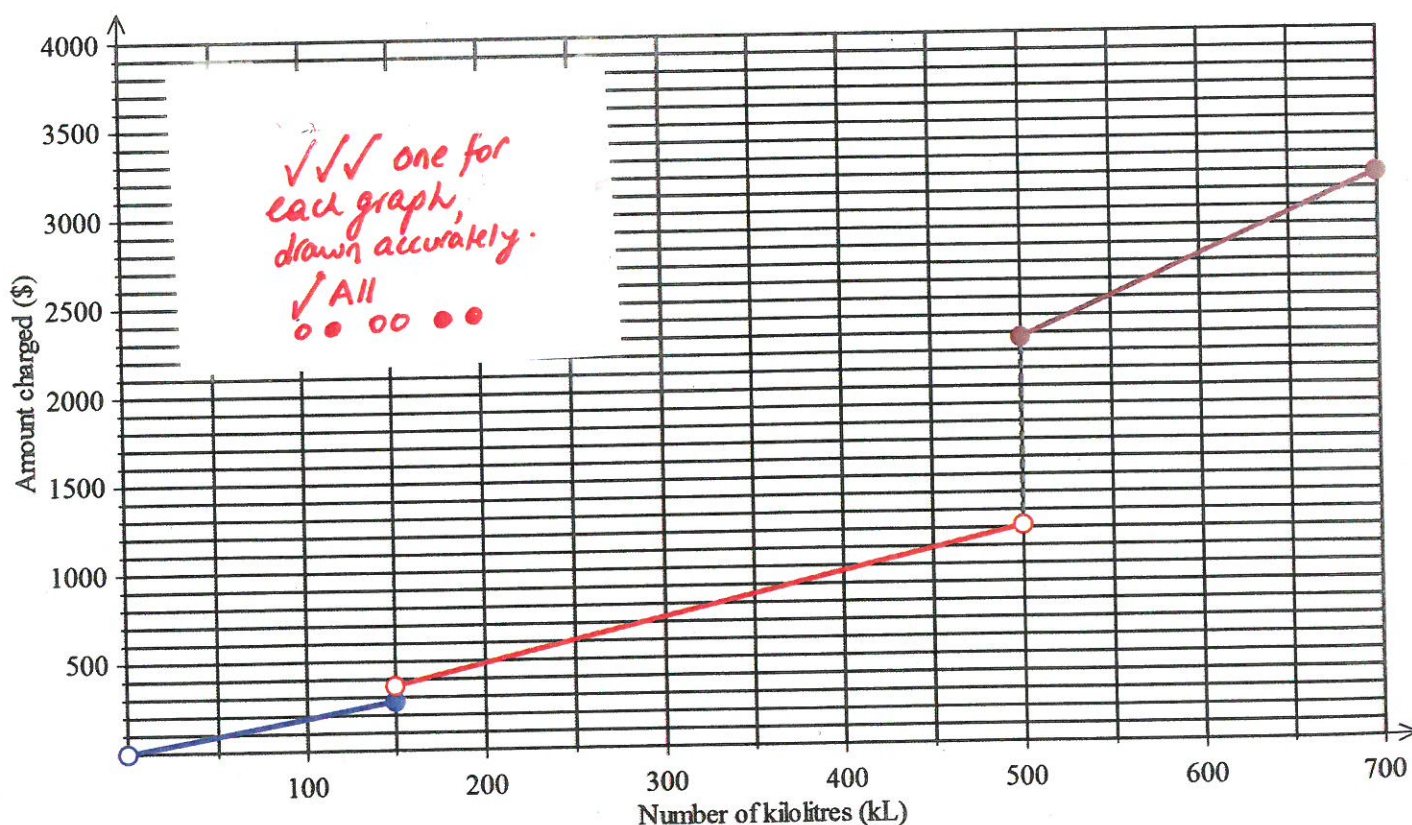
The Clean family have received their regular water account. Mrs Clean is shocked by the cost and vows to look a ways to save water. Water metres are read every two months which means every house in the state is visited six times every year.

Water use is calculated from the date of the last meter reading until the latest reading. The price of water per kilolitre increases which means the more water used, the more charged. The billing year begins with your first meter reading of the reading year, which can be any time from March to October.

For metropolitan areas, the pricing structure (approximated from the water corporation website) is tiered like this:



- (a) Use your calculator to assist you to represent this information on a suitable graph. Use the following set of axes to draw an appropriate graph to display the above pricing structure for up to 700 kilolitres of water use.



Across the billing year there are 6 billing periods. Water use accrues across the year. This explains why a water bill can get more expensive towards the last part of the billing year – even if less water than previous billing periods has been used. The cost per litre is increasing as you move into the higher water use bracket.

The Clean family's water bills are summarised in the following table.

- (b) Complete the final column of the table below. Remember the rate is calculated on the cumulative total. (for example, 70 kL is charged at the 150 kL rate as the total falls $(0 < x \leq 150)$).

Metre Reading	No of Kilolitres used in the two month period	Water use for the year. WATER is charged at this cumulative rate.
1	80 kL	80 kL
2	70 kL	80 kL + 70 kL = 150 kL
3	110 kL	260 kL
4	115 kL	375 kL ✓ calculates
5	125 kL	500 kL ✓ correct
6	85 kL	585 kL ✓ totals (F/T)

- (c) Complete the table below to show the Clean family's water charges for each period of the year and the Total charge.

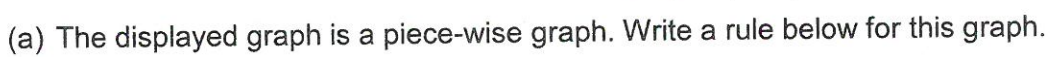
Metre Reading	No of Kilolitres used in two month period	Water Rate \$/kL	Water charge for two Month period.
1	80 kL	1.85	\$ 148
2	70 kL	1.85	\$ 129.50 ✓✓
3	110 kL	2.43	\$ 267.30 all six correctly
4	115 kL	2.43	\$ 279.45 calculated
5	125 kL	4.55	\$ 568.75 (-1 each error)
6	85 kL	4.55	\$ 386.75
TOTAL CHARGE			\$1779.75. ✓ correct total calculated (F/T)

- (d) Does your graph in (a) represent this situation? Explain

No, represents rates of each tier, but not cumulative calculations

✓ valid explanation

The graph below shows the fares charged by Optimus Bus Services for the first 25 km of travel.



4

- (b) Another bus service, Superior Bus Service, charges the following fares. Insert an appropriate graph on the above set of axes corresponding to the fares charged by First Bus Service.

$$C = \begin{cases} 80 & 0 < d \leq 5 \\ 120 & 5 < d \leq 10 \\ 150 & 10 < d \leq 15 \\ 190 & 15 < d \leq 20 \\ 210 & 20 < d \leq 25 \end{cases}$$

Graphs equations correctly.

✓ first step

✓ 2nd + 3rd step

✓ 4th + 5th step.

✓ 0 —● ALL shown correctly.

- (c) Show HOW to use the graph to determine the fare using the Superior Bus Route for an 18 km journey and state the fare.

\$ 1.90

✓ Shows answer clearly on graph
✓ Gives correct amount

- (d) Determine the difference in the fares of the two bus services if 12 km between Fremantle and Shenton Park was travelled.

14c

✓ Determines accurately
can get accurately from equations
 $C = 3(12) + 100$ and $C = 150$

- (e) Use the graphs to determine which Bus service charged the lower fare over the entire 25 km journey displayed. Explain your answer.

Superior Bus Cheaper over first 5 km ✓

Optimus Bus Cheaper between 5 km and $6\frac{2}{3}$ km ✓

Superior Bus cheaper between $6\frac{2}{3}$ km and 10 km. ✓

Optimus is cheaper for the rest of the 25 km journey. ✓

Clearly explains each of the 4 sections.

Accept 6.5 km for $6\frac{2}{3}$ km.
approx